

INTERNAL AND EXTERNAL VIRTUES

THEORIES are like apples; there are good ones and there are bad ones. A good apple is flavorful and nutritious. A good theory is one that we have good reason to believe is true, or at least likely to be true. Apples have all sorts of features that are indicative of goodness and that can be used at the store for making smart choices. An apple's color, shininess, shape, and firmness are related to its flavor and food value. Similarly, theories have features that are indicative of their truth, and the task of justification is to identify these features and use them to guide choices as to which theories to believe. In this way, responsible theory choice and responsible science are not unlike responsible shopping. The big difference is that you get to go home and eat the apple and find out if you were right and it really does taste good. With theories, however, there is no feature that is the taste of truth, at least not that we get to sample. There are only features that indicate truth. You never get to go home and see unambiguously that atomic theory, for example, is true. But you do the best you can.

Theories have lots of different properties and can differ one from another in a variety of different ways. A few have been mentioned already, namely, the degree of generality and hypothetical status of a theoretical claim. An account of the properties of a theory is an answer to the inquiry, "So tell me what this theory is like." There are plenty of things to say, that is, plenty of features to mention. There is the feature of how it was discovered, by whom, at what time of day. Some theories have the feature of being proposed by Einstein; some lack this feature. Some have

the feature of being liked by Einstein; some don't. There are also features of the content of a theory or of its form. Being logically consistent is a feature of form. Making reference to action at a distance, or to evil spirits, or to gluons are distinct features of content. There are even pragmatic features. Some theories are likely to be money-makers; others are not.

Clearly, some of these features of theories are irrelevant to an analysis of scientific knowledge and irrelevant to doing science. We want to focus on the important features, those that meet the following two criteria. First, they must be relevant to the likelihood of the theory being true. That is, they must be reliable indicators of truth so that our seeing whether the feature is present or not will be part of our warrant for believing the theory. Some theories have the property that you learned them on a Tuesday, but that is unimportant because it is neutral to their being true or false. Some theories have the property of describing goblin maintenance men to keep the world in good working order. That's an important feature because it helps us decide whether the theory is true or false.

The second criterion of importance is that the property is something we can in fact evaluate. The information must be accessible to us if it is to be of any use. To get back to apples for a moment, even if it's true that apples with darker seeds taste better, this isn't any help there in the store where information about the seeds isn't available. Similarly, a theory about the demise of dinosaurs is true if it is an accurate picture of the past events, but the feature of being an accurate picture is not something we can evaluate. We can't see what this theory is a picture of, and so we can't assess its accuracy. Much as we would like to use accuracy-of-picture or correspondence-to-facts as features that indicate truth, information about these features is unavailable. Features that are important to the justification of theories must be both truth-conducive and accessible.

The same standards are used in evaluating the testimony of a witness at a trial and in justifying a verdict. There are all sorts of features of a witness's testimony. It could be very loud or softly spoken, consistent or full of contradiction. Maybe it was given on a hot afternoon. Perhaps it is full of detail about the defendant. Some testimonies have the property of being fabricated in collusion with the defendant, and some, not necessarily others, have the property of being an accurate account of the crime. Note that some of these features of the testimony are what would be included in a reporter's description of the trial. Others, though, cannot be responsibly reported since their assessment relies on inaccessible information. We can't really tell whether the testimony has the property of being an accurate account of the crime, given that, in the context of the trial, the facts of the crime are not apparent. Responsible jurors must attend to those features of the testimony that can be evaluated with the available information, and of those, some will be helpful in determining the likelihood of whether the testimony is true or not. Start with the easy one, logical consistency. If the testimony contains contradictions, it can't be the whole truth and nothing but the truth. Then check for richness of detail. If the testimony is full of detail it is probably a report of the facts. Most people just aren't clever enough, or are too lazy, to fabricate their own details. The demeanor of the witness is an important feature too. A calm, confident witness who faces the jurors and the defendant during testimony is probably telling the truth, more likely, at least, than some shifty-eyed fidget who won't look you in the eye. These aspects of the testimony are more reliable indicators of its truth than are such features as being given on a hot afternoon, or on a Tuesday, or by a handsome witness or one who uses the word "really" altogether too often. These latter considerations are irrelevant.

The responsibility of the jury is to decide the truth of the criminal matter and to do this by finding and using the available,

truth-conducive features of the testimony. The responsibility of the scientist is much the same, to decide the truth of the matter about the natural or social world. The responsibility is carried out in a similar way, by finding and using the available, truth-conducive features of the theories. So too is the task of those who study science.

INTERNAL AND EXTERNAL FEATURES

It will be helpful in studying the features of theories to divide them into two groups. This will be just a preliminary distinction, subject to future changes, but it will be useful for identifying and evaluating the important features. Later on we will be forced to question the strictness and clarity of the dichotomy, though a more or less continuous spectrum may endure. It is not unnatural to understand a spectrum (such as temperature) by first understanding the features of its two ends (hot and cold).

Initially, then, features of theories can be grouped into those that are **internal** and those that are **external**. Internal features are those that can be evaluated without having to observe the world. That is, they don't depend on observations to check correlations between what the theory says and what the world is like. In this sense they are very economical since there is no need for experiments or laboratories to check for internal features of a theory. They can be evaluated by studying books, that is, by looking at the structure of the theory and its conceptual relation to other theories. Logical consistency is a clear example of an internal feature. Some theories have it; some theories don't. Checking for logical consistency is not an experimental procedure. If my theory of tomorrow's weather, for example, is that it will both rain all day and not rain all day, you don't have to look outdoors to realize that something is wrong. It's inconsistent, and this is evident from the internal structure of the theory.

External features, on the other hand, are relevant to the theory's relation to the world, and they require observation for their evaluation. For example, some theories have the property of making true predictions; some do not. It is impossible to tell whether a theory makes true predictions without actually doing the experiments and observing the outcome. It's not essential that you do these things yourself, but someone in the community of knowers must have a look to see how the experiments go if anyone is to claim responsibly of a theory that it makes predictions that come out right. Thus external features report contact with the world and therefore demand observational evaluation.

The courtroom-testimony analogy will help to clarify this distinction between internal and external features. Internal features are those properties of the testimony one could detect just by listening to the testimony and to that of other witnesses without checking the correlation of the testimony to objects or events in the world. If the witness says, for example, that on the night of the crime it was dark and raining heavily but says later under cross-examination that on the night of the crime the moon was visibly full, the testimony has the feature of being inconsistent. No one has to have seen the weather that night to realize that this is inconsistent. As another example of an internal feature of the testimony, suppose one witness says that it was very dark at the scene of the crime. This report then explains the account of another witness who claims she didn't see anyone in the back seat of the car. These two testimonies have a feature of explanatory coherence. They make sense together, and this is apparent just from hearing them. Again, no observation of the car or the scene of the crime is necessary to see that the two testimonies cohere.

Testimonies will also have external features, and that is why physical evidence and visits to the scene of the crime are important parts of a trial. If the witness describes the criminal as being skinny and having a mustache, then there is only one way to tell if this testimony has the property of being an accurate description

of the defendant, and that is to look at the defendant. Some member of the group responsible for deciding what's true, that is, some member of the jury, must perform the experiment and have a look.

The choice of terminology for this internal–external distinction may seem a bit mysterious as it is not obvious just what these features are supposed to be inside or outside. The internal features, since they do not speak of contact with the outside world are, in this sense, in the mind, at least the collective minds of the scientists or the jurors. They are discernible without the aid of the senses, our line of contact to the physical world. So, to evaluate the internal features of a theory is to think about it and see if it makes sense before you spend good time and money on the experiments to test it. To see if it makes sense, one compares the theory with other theories in mind, both in the individual's mind and in the collective mind of the group. External features, on the other hand, require information from the external world, from things other than ourselves. Evaluating these properties demands an influence from objects and events outside of our own minds and beliefs. Sometimes you even have to go literally outdoors and get dirty in order to check on external features. You never have to do this with internal features.

There are other pairs of terms that come close to describing this distinction, but none succeed quite as well as internal and external. We might consider, for example, speaking of theoretical virtues in distinction from experimental virtues. This is a close match to the internal–external distinction, but it is misleading in some important cases. Logical features, being free of contradictions, for example, do not fit neatly under the heading of being theoretical. They don't fit at all in the category of experimental. We could squeeze them in with the theoretical (we are, after all, in control of our own language), but then where do we squeeze the feature of making lots of predictions that will be amenable to observation? This is an internal feature because the fact that the

theory makes these predictions can be discerned by analyzing its structure and its implications. Whether or not these predictions come out to be true is another story. The point is that the feature of making predictions has the sound of being experimental, yet it is internal, indicating that the theoretical–experimental distinction could precipitate confusion over internal and external features.

Another potential expression of the relevant distinction between kinds of features focuses on the difference between features of coherence and features of correspondence. Philosophers often use these concepts to describe features of beliefs we have, whether or not the beliefs deal with issues of science. **Coherence** is descriptive of a whole system of beliefs and indicates that the constituents fit together in some way. It describes relations between one belief and other beliefs. **Correspondence** describes the relation between one belief and the world. It's one thing for my belief that the man next door is stealing my Sunday newspaper to fit nicely with my other beliefs that he is a cheap-skate and a crook; it's quite another for my belief to correspond to the fact of the matter as to whether or not he is stealing the newspapers. This distinction between coherence and correspondence is very similar to the separation of internal and external features, so close that we will at times drift into this terminology. The beliefs we have and that are analyzed by philosophers are just our theories about the everyday world, and questions about science are relevant to questions about knowledge in general. For now, though, we shall stick with the internal–external description. The idea of coherence implies an analysis of whole systems of theoretical claims to the exclusion of individual claims. We should leave open the possibility of internal features of individual theories.

The important, practical consideration in determining the value of any feature of a theory, whether it is internal or external, is whether it can be evaluated. We will want to use these features as

a measurement of how good the theory is. They will be the hallmarks of acceptability and so are of no value if they are hidden or incomprehensible.

VIRTUES

Like the members of the jury, scientists need to focus on the features of a theory that indicate that it is a good theory. These are virtues. These are the important factors for comparisons of theories, that is, for deciding between competing accounts of the world. There may not be an absolute scale of the goodness of a theory on which, for example, the heliocentric description of the solar system is a just-plain-good theory. But there ought to be a relative measure, a way of comparing theories, by which the heliocentric theory comes out as being better than the geocentric account.

Different aims of theorizing will work with different standards of goodness. There are various ways of answering the question, Good for what? If the aim of science, for example, is seen as keeping people safe from natural threats and from each other, then the theories ought to be chosen for their prudential virtues. These would be features of a theory that indicate that believing it won't lead to people getting hurt. Given this kind of goal, the responsibility of the sciences is a prudential responsibility, adopting theories that will not endanger people's safety. A prudential virtue might be that the theory is free of any descriptions of the release of large amounts of energy. For reasons of safety, that is, one might want to avoid a theory like $E = mc^2$. Conversely, it might be advisable to believe in a theory like the greenhouse effect and global warming, just to be on the safe side. If a theory predicts a preventable disaster, it makes good prudential sense to adopt the theory and implement the preventive measures, whether the theory is true or not. Better safe than sorry. Pascal was thinking along these lines when he decided to believe in the theory that

God exists. Think of it this way, he argued: Believing in God is safer than not, for if you believe in God and he exists, you have invested in a pleasant afterlife. If you believe in God and he doesn't exist, what's the loss? On the other hand, if you *don't* believe in God and he exists, you're in for trouble, and if he doesn't exist, you gain little in terms of making your life safe and pleasant. Clearly, predicting catastrophe for nonbelievers is a prudential virtue.

Of course, there are other standards by which to evaluate theories. Some of the features of theories will stand out as pragmatic virtues, properties that make the theory easier to work with or more useful. All other things being equal, for example, it is wiser to go with a simpler theory than with a more complicated one. If nothing else it will be easier to work with and less likely to incur mistakes in calculation or in reasoning. There are good practical reasons, in other words, to choose a simple, streamlined theory.

Theories may also have aesthetic virtues. Because of its symmetry or the elegance of its models, a theory could be appealing to the eye. These features just make it look good. The feature of being simple could be doing double duty here. It could be both a pragmatic and an aesthetic virtue.

There are also psychological virtues. If a theory has the property of giving us a feeling of understanding, of explaining something so that it now makes sense to us, it has psychological virtue. Astrological theories are like this in that, once you believe them, a lot of things are explained. My theory about the missing Sunday paper has similar virtue. I feel better now that I have an explanation for the missing newspapers. Before, it was so mysterious as to make me a bit uncomfortable.

All of these kinds of features of theories are nice, but they are not what we are interested in. Insofar as the aim of science is to deliver a true (though not necessarily safe, practical, beautiful, or satisfying) account of the world, it must focus on truth-conducive features. A theory with these features is more likely to be true

than a theory without them. These are the features we are interested in to answer the question, Why believe what science tells us about the world? It's not that the other kinds of virtues are simply unimportant. They are unimportant only for the particular goal of assessing the trustworthiness of science. Whether it's practical or not, safe or not, or beautiful or not, is it true? Any feature that supports an affirmative answer to this question is a truth-conducive virtue. From here on, truth-conducive virtues will be referred to as simply virtues.

Our project is to articulate the virtues. Within science, justification of a theory is the presentation of virtues. This is the process that wears away the label "hypothesis."

Some of the virtues mentioned earlier under other aims may serve as well to indicate truth. Simplicity, for example is a pragmatic virtue; it may also be a truth-conducive virtue. Being able to explain observed phenomena, for another example, is a psychological virtue; it too may be a truth-conducive virtue. We'll have to see. In other words, it remains to be demonstrated that these features of a theory indicate that the theory is likely to be true.

INTERNAL VIRTUES

Here is a description of important internal features of theories that are truth-conducive, that is, of internal virtues. This list is not above revision. There may be good reason in future to add to it or delete from it. It is, however, a good beginning for justifying theories.

1. Entrenchment

"Entrenchment" is not a very flattering name for a standard we regularly apply to assess the warrant to believe that a theory is

true. A theory is more believable if it is compatible with other, well-established beliefs about the world. In other words, a theory should be plausible, given what we already know. This requirement of initial plausibility is an impediment to wild and outlandish hypotheses that challenge the bulk of what is believed to be true of the world.

This property of entrenchment comes in degrees. A theory could fit in with other, background theories in the rather minimalist sense of simply being consistent with them. This could be pretty easy since a theory that talks about something entirely new will be consistent with all the old, entrenched theories no matter what it says. Where there is no contact there can be no contradiction. The theory of my neighbor the thief, for example, is perfectly consistent with the special theory of relativity, the theory of evolution, and all those other well-established beliefs about the world, and in that sense it fits in. It is a lonely fit, though, and a stricter level of entrenchment would have the theory be derivable from some other theories. The background knowledge should be not only compatible but also cooperative. In this sense, the fit, and the entrenchment, can be better or worse.

Entrenchment is a principle of conservatism, and it is clearly not a rule that is or should be followed with strict rigidity. Sometimes well-entrenched theories are refuted and wild and crazy theories turn out to be true. And sometimes calm, responsive, sincere witnesses are lying through their teeth. Einstein's special theory of relativity, claiming that measurements of mass and the passage of time depend on the motion of the observer, certainly lacks the feature of entrenchment, yet there is good warrant to believe the theory is true. Clearly, there are circumstances in which it is appropriate to relax or even ignore the standard of entrenchment. Clearly, or we would still think that everything is made of the elements earth, air, fire, and water.

Other things being equal, though, it is best to choose the theory that is compatible with the well-established background theories of the world. This way, the justification enjoyed by the entrenched beliefs is shared with the newcomer.

Entrenchment is the first of the virtues to be looked at closely, and it is the first indication that it is not going to be easy or straightforward to add up the score on a theory's virtues. The standard of entrenchment, like the other standards we will discuss, must be imposed with discretion, sometimes taken seriously but other times disregarded. The hard part will be in knowing, in the case of entrenchment, when conservatism is called for and when it is not. The answer will depend on the status of the other virtues. It must be all things considered.

2. Explanatory cooperation

Explanatory cooperation should really be an aspect of entrenchment, but the activity of explanation has a very high profile in science and is thereby deserving of its own status as a virtue. Explanatory cooperation means cooperation with other theories. It is the property of explaining, in part, why things are the way other theories say they are. Recall the example from the courtroom. Testimony that it was dark at the scene of the crime explains other testimony that no one was seen in the back seat of the car. The description of the crime starts to make sense now. The picture fits together and is therefore more likely to be true than if it were disjoint. In a scientific example, the theory that the boats used by common Greeks in the Bronze Age were small and unable to hold much cargo or travel far from the shore would explain, in part, why, as theory has it, fish was but a minimal part of diet in Bronze Age Greece. This explanatory cooperation makes both theories look good. It's not that justification flows from one to the other but that the explanatory link gives them mutual support.

This is different from explaining observed phenomena. It is different, for example, from the theory that fish was only a small part of the ancient diet explaining why only very few fish bones are found in excavations of Bronze Age sites. Being able to explain the observed phenomena is a very important feature for a theory to have, but it is an external feature. Its time will come.

3. Testability

Testability is a crucial aspect of science. A theory must make some predictions of observations that can be checked to see that the theory makes contact with the world. Sometimes this requirement of testability is identified as the essential ingredient that makes science scientific.

Testability is not really a truth-conducive virtue since being amenable to a test is different from passing the test. It is, though, a requirement for responsible justification of theories. A theory that lacks the feature of testability is precluded from evaluation for external virtues. That is, it can't have any external virtues. Because of its important role as a necessary prerequisite, the language can be stretched just a little to include testability as a virtue.

Being testable does not mean the same thing as being tested or as having passed tests. These features are clearly important, but they are external virtues and their time too will come. What testability does mean is that the theory is not insular and self-protective. It puts its credibility on the line by making predictions that are not guaranteed to be true, by being logical tautologies, say. This is a mark of responsibility. If the theory is false it will be exposed as false. Without verifiable predictions, the theory is left to its own proclamations of truth just as a witness on the stand who says nothing that can be independently checked must be taken at her word.

Sometimes the feature of testability is described using the concept of falsifiability or refutability. This is, of course, distinct from being falsified or refuted. It means simply making predictions that are not predetermined to be right, predictions whose outcome depends on how the world is. In this way, the testing of the predictions will be meaningful and is more likely to expose weaknesses of the theory. Again, if the theory is false, this is how we will find out.

The more precise a prediction, the more informative will be the testing, and so precision is an aspect of testability. A vague prediction could too easily be counted as true even if the theory that sponsors it is false. The smaller the target, that is, the more one's skill is tested. Theories of astrology tend to fail on this evaluation. Based on the arrangement of stars and planets at someone's birth, the theory predicts such things as a tendency toward generosity or a tendency toward seriousness. How can you test things so nebulous as tendencies and so imprecise as seriousness? Hitting such large and unclear targets is no real test of the accuracy of the theory. Testability, in other words, requires not only observable predictions but precise predictions as well. If I propose a theory, for one more brief example, that predicts that some sort of earthquake will occur sometime in the next hundred years, the prediction is observable and will quite likely be true. It won't however, be much of a test of the theory. If this is the extent of its predicting, the theory is not falsifiable enough to be taken seriously.

In sum, then, testability is an important prerequisite for a good scientific theory. It is a feature of the structure and content of the theory which indicates that it is amenable to testing against the observations. It is an internal virtue which is necessary for there to be any external virtues. The theory must first make predictions and explanations before these can be compared with observation. The predictions must be neither predetermined, as are logical

and mathematical claims, nor indeterminate, as are very vague claims. How the theory fares in its contact with observations, that is, whether the precise predictions are true or false, is then an external feature.

4. Generality

General claims about the world are more interesting than specific claims, but that doesn't mean they are more likely to be true. There is, in fact, a kind of trade-off in that the more sweeping a statement, the more risk it takes. On the other hand, if there is some basic unity to the world, some small number of fundamental parts and operating principles, then only very general theories can capture these basic truths. With the assumption of unity, the feature of generality is a truth-conducive virtue. Without the assumption, the values of generality are pragmatic, aesthetic, and psychological in that generality is an aid in organizing our understanding.

Besides being a clue that a theory is getting at the deeper truths about the world, generality is a valuable feature because it enhances a theory's testability. If the theory generalizes over time such that its description is applicable at all (or many) times, then the theory is amenable to repeatable tests. Relevant experiments can be performed more than once with the expectation of the same results, thus making the tests more reliable and more accountable. This is a virtue. If a theory generalizes over many kinds of things, then it is amenable to a variety of tests. The greater diversity in the testing sample, the more at risk is the theory and hence the more significant is the testing. Generality, in the sense of describing a large variety of things, enhances falsifiability. If, for example, a theory describes some aspect of the characteristics of all electrons and not just the ones in this room or just the ones in hydrogen atoms, then the theory

has the obligation to predict correctly the electron-related behavior in all conditions, not just the limited circumstances in this room or in hydrogen. The theory is in this sense more open to testing.

5. Simplicity

Simplicity is a popular choice for an internal virtue of a theory. A good theory should be simple, or, at least, the simpler the theory the better. This approach to theory assessment is sometimes referred to as using Ockham's razor, after the medieval philosopher whose methodological advice was to shave theory clean of all unnecessary complication. Make it as simple and uncluttered as possible. Not only is the streamlined theory more likely to be true; it is also much easier to work with and more rewardingly elegant to present.

There are two points to clear up before simplicity can be used as a truth-conducive virtue. They are exactly the two questions that must be asked of any feature that is cited as indicating that a theory is true: How can the feature be evaluated? Does it really indicate that the theory is likely to be true?

How can we tell that a theory is simple, or how can we compare two theories to determine which is simpler? Sometimes it's easy. If the theory is mathematical, it can be rather straightforward to assess the degree of complication of the equations, whether they are linear or quadratic functions or perhaps exponentials and hyperbolic cosines. Also relevant to the measure of simplicity of a mathematical statement is the number of variables and constants. The more of these participants needed in the formula, the more features needed in the description of the world, and hence the world seems more complicated.

Even without the formulas and equations, it is easy to see how mathematical simplicity can be determined and used as a factor in theory choice. Consider the example of a scientific experiment

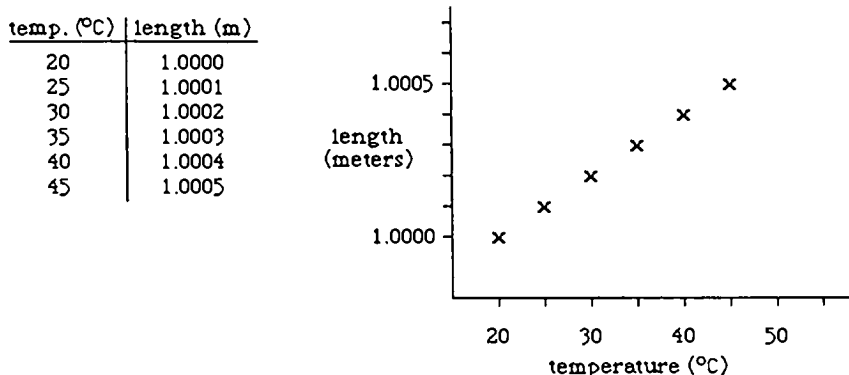


Figure 2.1. The data of measuring the length of a brass bar at various temperatures, presented on both a table (*left*) and a graph (*right*).

to understand the relationship between the temperature of a brass bar and its length. At room temperature (20° C) the bar is one meter long (1.000 meters, to be precise). The bar is heated and its length is measured at five-degree temperature intervals, that is, at temperature 25° C, again at 30° C, and so on. The results of the length-versus-temperature measurement are tabulated and plotted in Figure 2.1. At this point the observed phenomenon has been carefully described. Theorizing begins with the projection of the general relation between length and temperature, that is, the description of what the length would be at any temperature, whether it's one of the ones measured or above these or in-between. A theory would specify how to draw a line on the graph that passes through the measured points and extends the information to intermediate and higher-temperature points. Given the finite number of measurements, the finite number of data points on the graph, that this and every other experiment are limited to, there will be lots of lines that pass through the data points. There will be lots of theories that are compatible with the observations. These theories have equal scores with respect to external virtues, and this is exactly why internal virtues are so important. If an appeal to the observations is indecisive between

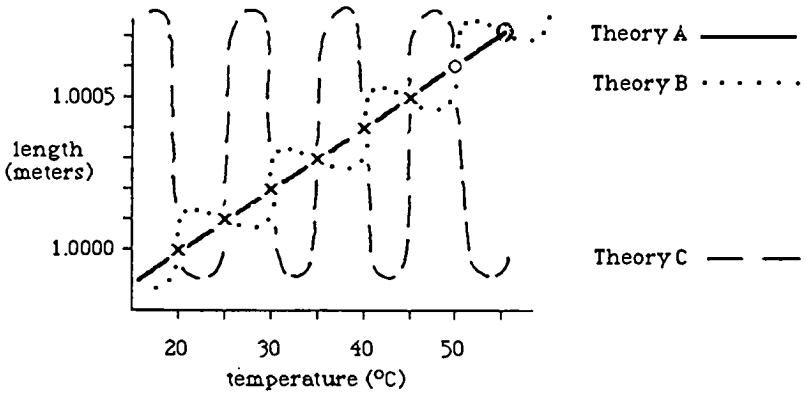


Figure 2.2. Three possible theories, A, B, and C, which agree on all the data, both past (x) and future (o), but disagree on the properties of the brass bar.

theories, the choice between the theories must be made not simply on the basis that they account for the phenomena but on *how* they account for the phenomena.

With the example of the heated bar, a few of the many possible theories, all of which are compatible with the data, are sketched in Figure 2.2. The points on the graph don't decide between theories A, B, and C. Not even tests of predictions of what the length will be when the temperature is 50° C will be decisive. But an appeal to simplicity rules in favor of theory A, the straight line. As Ockham might say, why think that the unmeasured lengths oscillate wildly back and forth when the length we chose to measure increases so clearly in a straight line? In this case it is easy to see which is the simplest theory, and it is easy to see that its simplicity is counted as a virtue.

Usually, though, evaluating simplicity is not this easy. Interesting theories, those that are challenging and important to justify, are never just lines through data points on a graph. What about the theory of evolution, for example? Is it simple? More to the point of decision making, is the theory of evolution simpler than the theory of special creation? It's not just that this question is

difficult to answer. It's difficult to know how to begin the evaluation to come up with an answer. And this is typical in assessing the simplicity of theories. It's unclear, to cite another example, that Einstein's special theory of relativity, with its description of time and mass as being dependent on the observer's motion, is simpler than Newton's mechanical account of the world.

Furthermore, the measure of simplicity of a theory will be influenced by the language and concepts used by the evaluator to describe the world. Different languages and theories will describe the same situation with different numbers of things. For example, there is a large variety of elemental building blocks in the world. There are hydrogen, helium, carbon, xenon, and so on. In the context of the atomic theory, though, these are all the same kind of thing. They are all atoms. Employment of the atomic language, then, shrinks the number of kinds of entities to one, and if simplicity is to be judged on the basis of number of entities, it will be a feature of the choice of language rather than the content of a theory. The measure of simplicity, in other words, is not an independent, objective standard but is dependent on the linguistic and theoretical context. This is true even in the mathematical case. The appearance of the graphed data and theory, as well as the degree of complication of the descriptive formula, depends on the coordinate system used, that is, on the choice of mathematical language. The straight line in the heated-bar example is the simple result of our choice of Cartesian coordinates. Plot the same data in polar coordinates and the simplicity disappears.

Simplicity, in other words, is not an unambiguous or easy thing to evaluate. It is sensitive to the choice of language of description instead of being stably anchored to the content of the theory. This not only makes it hard to assess but makes its link to truth questionable.

What does simplicity have to do with truth? Worries aside about the difficulties of evaluation, why should simplicity in a theory be taken as a sign that the theory is likely to be true? This

is not to suggest that simplicity is not a truth-conducive virtue, but that claim does bear the burden of proof. Simplicity is clearly a pragmatic virtue, and for that reason it is a good thing to strive for. But we have yet to see the connection between being simple and being true.

This quick survey of some of the important and popular internal virtues has shown that none of them can be evaluated in a straightforward way. The answer we get as to how virtuous is a theory depends on our other theoretical preconceptions. This is clearly true of entrenchment and explanatory cooperation, and it is true as well of simplicity. Different people, with different beliefs about the world, will see different theories as having different internal virtues. This clearly limits the value of these virtues as objective criteria for justifying and choosing theories. It indicates, as well, an aspect of the self-regulation of science mentioned in the Introduction. If these standards of justification depend on the background language and theories of science, then the application of the standards is a task for the experts who speak the language and understand the theories. It is not just a scientist–nonscientist separation that is important here. The users and believers of theory *A* may honestly see their claims as simpler while the users and believers of theory *B* judge their own claims as simpler. If the judgment on virtues is theory-dependent, it seems the virtues cannot always be decisive between theories.

As further complication in the use of internal virtues to justify theories, note that there will always be conflicts among the virtues of a particular theory, and trade-offs must be made. There will be conflicts among internal virtues and conflicts with external virtues. Simplicity, for example, conflicts with generality. The speculative new theory of superstrings, which unifies the quantum descriptions of matter and the relativistic description of gravity, thus tipping the scale of generality, does so at the expense of describing the events of the world in twenty-six dimensions in-

stead of the usual four (three spatial dimensions and time). The theory does well on generality but poorly on simplicity, and, indeed, no theory will win across the board. Nor is there any indication of which virtues are more important. There is no hierarchy to rank the importance of virtues so that if one infringes on another it is clear which has the higher authority. In this sense, the list of internal virtues is like the Bill of Rights. Neither specifies how to resolve conflicts among various rights or values, and so both must be administered with discretion.

This discretion involves the recognition that even though these features of theories are virtues they must sometimes be compromised. It may be justified to believe a theory that lacks particular virtues, justified because the theory is better on other virtues, possibly including external virtues. Obviously, it is unwise to be a slave to entrenchment, as this would stifle any change in science. It is nice, and probably justified, to think that some progress has been made since the theory of four elements, earth, air, fire, and water. Nor is it advisable to depend single-mindedly on simplicity, lest science become simpleminded. There are trade-offs to be made, both for other internal virtues and for external virtues.

EXTERNAL VIRTUES

External virtues are very important. They must be a heavily weighted part of the total score in justifying a theory. These are the features, after all, that indicate that the theory is compatible with experimental results, that is, that the theory is accountable to observations. These virtues will be only briefly described now because their importance warrants separate chapters.

1. Explanation

A good theory should be able to explain some observed phenomena. For example, the coincidence between the ocean tides and

the position and phase of the moon has been apparent for a long time. Newton's theory of universal gravity explains this coincidence by pointing out that the situation is an instance of a general law. The objects are the kinds of things that gravitationally pull on each other, and so it is to be expected that the pull of the moon would cause the water in the ocean to move. The explanation shows how the particular phenomenon fits in the general pictures of things, and this fit makes the picture seem accurate.

Even better, of course, is for the theory to explain lots of phenomena. The more pieces of the picture that can be shown to fit, the bigger the sample of explained phenomena, the more likely it is that the theory is true. Better still, a good theory should explain a wide variety of stuff. Instead of building a large sample by doing the same test over and over, a theory's truth is put more on the line by a broad sampling of tests. The fact that Newton's theory can explain not only the tides but also the orbits of the planets and the motion of falling objects weighs heavily in its favor, more so than the fact that it explains the tide on Monday, Tuesday, Wednesday, and so on.

It is important to note that there are other theories that explain the tides. The general theory of relativity is one, and I'm sure we could find or invent plenty of others. Here is one in the invented column. Maybe there are beautiful but shy sirens who live deep in the sea, sirens whose great love of the moon causes them to rise from the water when the moon is overhead. They rise and part the seas, forcing the water to swell up on the shores, thus explaining the relation between the moon and the tides. It's their shyness that has kept the sirens from being seen.

This is a stupid theory, and it does not warrant anyone's belief. But its failure cannot be blamed on a simple lack of external virtues. It does explain the tides, and with sufficient fixing up it could accommodate lots of other phenomena. It could be made compatible with the data. It must be excluded, then, on the

basis of internal virtues. For example, it is not a very testable theory in that it can blame any failure to see the sirens on their convenient shyness.

The moral of the story is this: As important as it is for a theory to explain the phenomena, being an explanation does not decide unequivocally in favor of a theory. It has to be a good explanation, even the best explanation, and this measure of goodness is an internal feature.

2. Testing and confirmation

Testing is not strictly distinct from explanation, since it too is done by drawing observational consequences from the theory and seeing that they fit the facts. The difference is that testing is usually accomplished by looking at new evidence as predicted by the theory, whereas explanation is usually of phenomena already observed. The emphasis in testing is not so much on *why* something happens as simply on the fact *that* it happens according to prediction. Newton's theory of gravity, for example, not only explains the observed coincidence of the tides and the position of the moon but can predict occurrences of eclipses. These precise predictions are exactly what counts toward falsifiability. If an eclipse occurs as predicted, that is a virtue of the theory and the event is counted toward confirmation or corroboration of the theory. If the eclipse does not occur, that's a flaw of the theory and the event is counted toward falsification or refutation.

This scheme accurately captures the basics, but only the basics, of the testing of theories in science today. The current theory of the sun, for example, describes a furnace of nuclear reactions in the solar interior, producing, among other things, a lot of neutrinos. The theory predicts, then, that some of the neutrinos ought to be detectable on the earth. A productive step toward answering the question "Is this theory true?" is to look for the neutrinos,

that is, to see if the prediction is true. This is an experiment in progress, and we will hear more about it later, not only in this book but in the science news.

WHAT'S NEXT?

The external virtues warrant and will get a much more careful analysis than they got in this chapter. They deserve this care because they are so important. Accountability to observation is much of what intuition says is scientific about science. They deserve a deeper look also because they are significantly more complicated than has been indicated so far.

The closer look at external virtues of theories will eventually lead to the demonstration that the dichotomy between internal and external features is untenable. What look like external features, it will turn out, really aren't. Insofar as features are relevant to justifying theories, they are internal. What will happen, then, is that the lists of internal and external virtues will be merged and expanded. In all discussions of virtues, it is important to be keen to the following questions: How can the feature be evaluated? Does it indicate that the theory is likely to be true? How is it to be used with discretion, and how are we to make trade-offs if this virtue conflicts with others?